

AgentWorkShop: A Node-Native Integration Framework for LLM Multi-Agent Closed-Loop Control of Heterogeneous Industrial Production Lines

First Author, Second Author, and Third Author

Abstract—Industrial LLM-agent deployments stop at the advisory boundary: published systems diagnose, generate PLC code, or recommend setpoints; to our knowledge, none integrates an agent team into a production line’s own protocol, data, and semantic fabric. We present AgentWorkShop, a configuration-driven integration framework that embeds LLM multi-agent teams natively into the supervisory fabric of heterogeneous production lines: a multi-harness agent runtime (14 interchangeable engines) and an industrial runtime (five protocol drivers with per-node edge acquisition) execute as one process over one data domain. Commissioning is configuration, not code: one plant-scenario identifier re-instantiates the deployment on a second, independently defined line with zero platform code changes; across both scenarios, 33 seeded out-of-window/range writes are intercepted (Wilson 95 % [89.6 %, 100 %]) with zero false blocks. Every agent-issued write traverses one governed path—recipe-window interlock, mode-bound approval, readback-verified write, attributable journal—so the accepted-action language is constrained by construction, and the backstop’s automatic rollback is exercised end-to-end on a live breach. All plants are seeded simulators. A governed closed-loop benchmark on a cast-film twin recovers 96.5–97.3 % of the offline grid optimum in two iterations and at most two governed writes per seed under a fixed controller (quoted run: 97.0–97.2 %); four released LLM-agent-team campaigns closed the same loop over real transports with every write attributed, but reached only a median 0.76 of that optimum—a measured shortfall that motivates the framework’s propose/dispose design. All results come from one executable pipeline with machine-checkable reproduction against frozen baselines.

Index Terms—Agent integration framework, closed-loop control, digital twin, human-in-the-loop, industrial automation, large language models, multi-agent systems, node-native integration, protocol heterogeneity.

I. INTRODUCTION

INDUSTRIAL enterprises are deploying large language model (LLM) agents widely [1], [2]. Yet essentially all published industrial LLM systems stop at the *advisory boundary*: multi-agent frameworks build industrial knowledge graphs [3], generate and verify PLC code [4], [5], and parameterize digital-twin simulations [6], [7]; beyond single-stack control interfaces [8] and simulated fault-tolerant control [9], in none does an agent team’s decision become a governed write on

a protocol-heterogeneous plant—all share an *integration posture*: the agent advises a human or a compiler, never joining the plant’s own protocol, data, and semantic fabric. The boundary is well motivated—an unguarded write path from a stochastic model to physical actuators is unacceptable in production—but it has left the central question of agentic industrial automation unanswered quantitatively: *what does it cost, in integration effort, latency, and human attention, to let an LLM agent team close the control loop on a real plant, and what safety does the resulting governance buy?* This paper measures the first two costs and contributes the human-attention arm as a released protocol design.

Answering it requires three things no existing work provides together: (i) a *node-native integration substrate*—agents, protocol drivers, and plant data in one process and one data domain, so binding an agent to an actuator is a configuration act, not an integration project; (ii) a *governed write path* whose accepted-action language is safe by construction, attributable, recoverable, and formalized enough to be checked rather than asserted; and (iii) a *reproducible evaluation protocol* measuring integration adaptability and the write path’s safety—overhead trade-off across protocols, plant scenarios, and fault conditions.

We present AGENTWORKSHOP, a configuration-driven *integration framework* that meets all three requirements, together with its evaluation suite, AW-INDUSTRIALBENCH. It couples an *agent half* (14 interchangeable LLM harnesses, lead-based team supervision, episodic–semantic memory) with an *industrial half* (five protocol drivers, per-node edge acquisition runtimes, a hysteresis-and-debounce alarm chain, and time-series/frame storage); the halves meet at a *governed bridge*. Nodes are bound to agents, physical semantics are injected as semantic cards, and the agent tool surface exposes exactly one write path—the *governed write-control pipeline*: a batch-scoped recipe-window *interlock*, a mode-bound human-approval *gate*, a readback-verified *write*, and a signed, attributable *journal*. A system-level *backstop* scores open optimization records against the recipe window and executes bounded rollbacks. Because the agent runtime and the node drivers share one process, this contract is *enforceable* rather than merely declared. Commissioning is configuration, not code: driver configurations derive at runtime from each device’s integration export, and a named *plant scenario* re-instantiates the full deployment; Sec. VI-B recommissions the

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same live instance onto a second scenario by switching one identifier, with zero platform code changes.

Our contributions are:

- 1) **A node-native integration framework** for LLM multi-agent teams on heterogeneous production lines: agent and industrial runtimes in one process and one data domain; five protocol drivers with per-node edge acquisition; physical-semantic grounding via semantic cards; and team-level node binding under a 14-engine harness registry, keeping agent intelligence replaceable (Sec. III).
- 2) **A governed write-control pipeline** for agent-issued actuator writes, formalized as a supervised state machine with three safety invariants (containment, readback closure, bounded autonomy), each mapped to implementation anchors and negative acceptance tests (Sec. IV).
- 3) **AW-INDUSTRIALBENCH**, a reproducible evaluation protocol with a quantified safety-overhead design: to our knowledge the first experimental design measuring the interception rate, false-block rate, and write-latency distribution of an approval-gated agent write path, with the human-attention dimension fixed by protocol as a human-operator arm (E8, Sec. V), against governance ablations, general frameworks, human operators, classical control, and cross-engine replications; the governance ablation is measured here, the rest are released designs. The measured arm (Secs. VI-A–VI-G) shows the cost is structural, not temporal: bypassing the soft interlock drops interception from 100 % to 66.7 %—breach writes execute and are journaled—with no resolvable latency added.
- 4) **A completed closed-loop program on a seeded extrusion cast-film line**: a governed controller recovers $J/J^* = 97.0\text{--}97.2\%$ of the offline grid optimum ($W^*=89.894$) in two iterations and at most two governed writes per plant seed across three seeds, and four released LLM-agent-team campaigns close the same loop over real transports with every write attributed but reach only a median 0.76 of that optimum—a measured negative that motivates the propose/dispose split (Secs. VI-F–VI-H).

II. RELATED WORK

A. Advisory-Boundary Systems

Industrial LLM applications cluster into advisory families. *Diagnosis and assistance*: CoMA-IKG builds industrial knowledge graphs with LLM agents [3]; surveys chart the breadth [1]. *Engineering automation* generates control code: Agents4PLC verifies PLC code against formal specifications [4]; MetaIndux-PLC adds control-logic guidance [5]. *DT-coupled agents* parameterize simulations [6] and use a twin as prompt source and validation sandbox [7]. All share two boundaries: output reaches a human or a compiler, never an actuator; and the DT twin is the *destination* of agent action, whereas in AGENTWORKSHOP it merely observes a real protocol stack—the destination is the plant. *Closed-loop LLM control*, the newest family, removes the boundary itself: LLM agents have turned fault detections into fault-tolerant control

actions on simulated plants [9], and tool-equipped LLMs have controlled real machines over OPC UA in natural language [8]. LLM agents also actuate equipment in adjacent domains—autonomous chemistry [10] and robot control [11]—but target single-apparatus loops, not production lines. These works couple one agent (or team) to one protocol stack with per-application governance; none contributes a reusable integration substrate (protocol-heterogeneous, node-native, team-level binding) or a reproducible governed-write benchmark—the gap AGENTWORKSHOP fills. The alarm chain of Sec. III-B follows alarm-management practice (ISA-18.2) [12].

B. Governed Execution, Benchmarks, and Frameworks

Outside industry, τ -bench [13] supplies our approval-gate evaluation paradigm. Hard action enforcement—control barrier functions [14], shielding [15]—has its own line; here the wrapped component is an *agent tool surface*, whose fallback path (interlock, approval, readback, journal) is part of the audited contract. Approval semantics rest on levels-of-automation research [16]. The line/product/recipe/run tree of Sec. III-F combines ISA-95 [17] with the ISA-88 recipe/batch model [18]; the semantic card is the node-level counterpart of the Asset Administration Shell (IEC 63278) [19]. General agent benchmarks [20] involve no physical plant or actuator safety contract; industrial datasets (TEP [21], SWaT [22]) target fault diagnosis, not closed-loop control—AW-INDUSTRIALBENCH bridges the two. Orchestration frameworks (AutoGen [23], LangGraph [24], CrewAI [25]; on ReAct [26]) provide scheduling but no protocol layer, semantic grounding, or write governance; AW-INDUSTRIALBENCH treats them as baselines over the same MCP tool surface [27]. Network segmentation follows IEC 62443 [28], complementing IEC 61511’s basic-control/safety separation [29]; AGENTWORKSHOP applies it at the agent-plant layer. The gap AW-INDUSTRIALBENCH occupies is node-level agent integration combined with closed-loop actuation under a formalized governance pipeline and per-write attribution.

III. SYSTEM ARCHITECTURE

AGENTWORKSHOP (v0.7.39) is an integration framework, not an application: in one Node.js process it ships the two halves that industrial deployments otherwise bolt together by hand—a REST and WebSocket API, an MCP server with 25 tools [27], an A2A agent interface [30], and a hot-reloading, error-isolated plugin host—under one authentication layer. Operators enter through a web console or TUI workbench sharing the same approval queue. Fig. 1 shows the complete architecture.

A. Design Principles

PI, node-native integration: the agent and industrial runtimes execute in one process and one data domain, so binding, telemetry, and write control share invariants rather than APIs—hence *check-then-actuate integrity*: the interlock test and the driver call run in one event-loop turn over the same process-local recipe-run state, and the window that authorizes

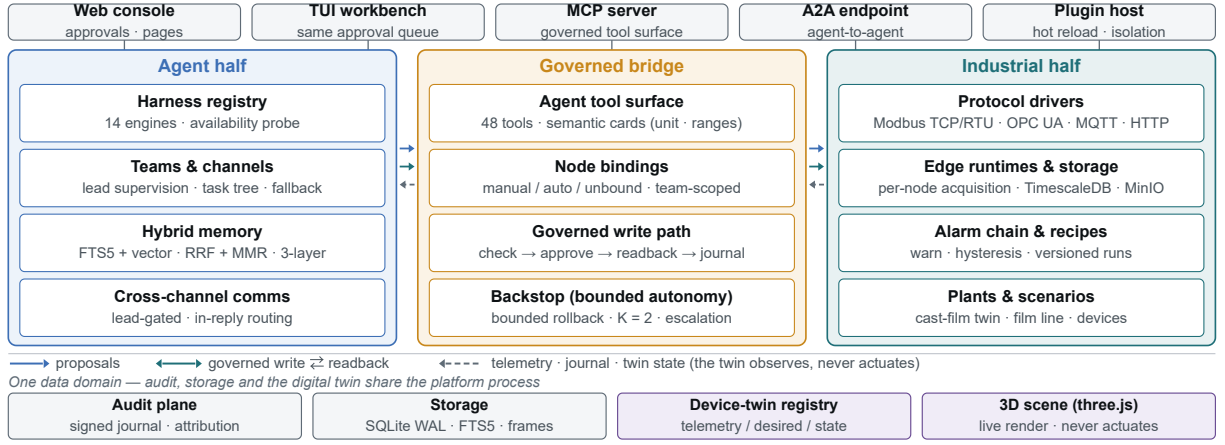


Fig. 1. AGENTWORKSHOP integration framework. **Left (agent half):** 14-engine harness registry, lead-supervised teams, episodic-semantic memory. **Center (governed bridge, amber):** the only path from agent intent to actuation—semantic cards, node bindings, Algorithm 1’s governed write path, the bounded-autonomy backstop. **Right (industrial half):** five protocol drivers, edge runtimes, starvation-free gateway sweep, alarm chain, line/recipe management; connectivity ends at real devices or the two seeded plant scenarios. **Top (entry surfaces):** web console, TUI workbench, MCP server, A2A endpoint, plugin host—each dropping into the same platform process. **Bottom (one data domain):** audit journal, storage, device-twin registry feeding the three.js scene; the twin renders live state but never actuates.

a write cannot change between check and actuation. Co-residency governs the platform-internal path but does not contain the harness engines, which remain privileged local processes (containment is a deployment property; Sec. IV-D). *P2, supervisory positioning:* AGENTWORKSHOP occupies the supervisory, soft-real-time tier; hard real-time control and safety functions remain in PLCs, mirroring the IEC 61511 separation of basic control from safety instrumented functions [29]. *P3, configuration-driven, down to the plant scenario:* a typed configuration schema drives acquisition intervals, approval policies, retention, and memory budgets; driver configurations are derived from each device’s integration export, and a named *plant scenario*—devices, signal semantics, dynamics, recipes—re-instantiates the full deployment, so adapting to a new line is a configuration task (measured in Sec. VI-B). *P4, engine-agnostic governance:* agent intelligence is a replaceable component; the governance pipeline of Sec. IV is defined at the tool boundary and is therefore identical for every engine.

B. Industrial Half

Five protocol drivers spanning four protocol families—Modbus TCP, Modbus RTU (over TCP), OPC UA [31], MQTT, and HTTP/REST—sit behind a uniform, plugin-registerable driver interface; a mock driver supports deterministic testing. Each acquisition node runs an *edge runtime* owning its sampling schedule and in-flight mutual exclusion, with three decoupled intervals: acquisition (default 5 s, floor 1 s, hard clamp [0.1, 60] s; sub-tick intervals wait for the 250 ms sweep), WebSocket publish (default 1 s), and display/trend (default 5 s). A 250 ms gateway sweep samples nodes under a non-accumulating budget of 64 in-flight samples per tick; a *rotating cursor* plus per-node *due check* stops not-yet-due nodes from consuming budget first, so tail nodes are never starved.

Each node’s alarm state machine spans ok/warn/alarm/offline: 8 % warn band, 2 % hysteresis,

three-frame ok↔warn debounce, watermark-based out-of-order defense, recipe-window violations as edge-triggered alarm; acknowledgment is a human step with escalation (default 15 min). Time series persist to TimescaleDB (batched 500 ms flushes, tiered retention), frames to MinIO with fallback to local disk, and platform state to SQLite (WAL, FTS5); every alarm, write, recipe change, and rollback row carries line, product, and recipe dimensions for Sec. VI’s metric slices.

C. Agent Half

Fourteen engines are registered with a capability matrix (mid-task steering, supervision, human-in-the-loop, among others): an in-process deterministic mock (used in Sec. VI), a first-party RPC engine (omp), and vendor or open-source CLI/SDK agents (e.g., codex, claude, qwen). Capabilities are asserted per engine and verified by the static audit (Sec. VI-C); dispatch follows an allow-list probe that resolves each executable and validates its response.

The collaboration unit is the *channel*: a named room hosting one lead, its workers, the task tree, and the message history; teams deploy as a unit, and every industrial binding is scoped to its channel. Task trees carry seven states (SUBMITTED, ASSIGNED, WORKING, WAITING for blocked parents, FAILED, CANCELLED, COMPLETED). Each lead runs a supervision loop on a 1 s tick with exponential idle backoff and event wake-up; each tick first assembles a state fingerprint (task tree, member queue depths, newest mailbox message) and skips the LLM call if unchanged—an idle plant costs no tokens unless a parent task is closeable. When consulted, the LLM owns goal judgment, completion, and escalation; mechanical decisions run every tick on a deterministic rule engine: FIFO dispatch to the worker with the shortest queue (ties broken by longest idle), reassignment of failed tasks while retries remain (limit 3), and a 300 s stall watchdog treating tool activity as

liveness. Both are ablated as E2/E3 (i.e., B6/B7), and their constants anchor the static audit (Sec. VI-C).

Agents accumulate typed memories (session, task, peer, team chronicle, reflection, semantic) in SQLite with FTS5 and an optional vector index. Retrieval fuses a lexical channel (FTS5) and a dense vector channel by reciprocal-rank fusion [32] ($k=60$), scores candidates by $0.5\rho + 0.3\phi + 0.2\iota$ (normalized relevance, kind-aware recency with 3–14 d half-lives, access-weighted importance), diversifies by maximal-marginal-relevance [33] ($\lambda=0.7$), and injects three layers under a 500-token budget. Compaction triggers at 70% of the context window and harvests summaries back into memory.

D. The Governed Bridge

An engineer binds plant nodes to agents (single agent or whole team); the binding injects physical semantics into the agents' context and derives their tool surface—no separate integration layer to build or maintain. A bound node yields a *semantic card*: physical quantity, unit and display precision, *hard interlock* range, *soft interlock* recipe window, warn band, live value with freshness, and a rule limiting any single step to 2% of span. Bindings carry an operating mode: *manual* (every agent write approval-gated), *auto* (writes proceed under interlock and backstop), or *unbound* (no write path); team members can share one node set, so a lead-dispatched task and a worker's actuation share one authorization scope. Agents never see register addresses.

Agents act through a host-tool surface of 48 tools, 23 of them industrial or automated-ML (the MCP server exposes a 25-tool subset), spanning setpoint control (`dcw_control`), reading, judgment, rollback, audit, and recipe editing under approval. Two separations are architectural: *judgment is not execution*—`dcw_judge` records a keep/rollback/uncertain verdict but cannot actuate—and *lead-only tools* are restricted to lead agents with approval.

E. Digital Twin and Node-Binding Rendering

Rendering binding is part of the integration model: a *device-twin registry* persists, per scene entity, the bound 3D asset, the agent in charge (a virtual operator avatar), and the command set, alongside the three device-shadow state planes [34] (*telemetry*, *desired*, *state*). A WebSocket bus broadcasts lifecycle events and every reading, so the scene renders live values without polling; scene edits write back to the node registry, so the scene doubles as a commissioning editor that never actuates—*desired-to-state* reconciliation traverses the governed write path when it does (Sec. IV).

F. Production-Line Management

The framework models a production line as a four-level object tree—line, product, recipe, *run*—because every governance object of Sec. IV hangs off it: the recipe window constrains writes jointly with the hard range while a run is active (Eq. (1)), acquisition samples only nodes on lines with an active batch (PLC-side protection persists between batches per P2), and the backstop scores open records against the active recipe window (Sec. IV-B). The versioned, non-destructive

recipe lifecycle spans editing (new version $n+1$), *known-good* baseline freezing, *known-good rollback* that re-dispatches the frozen set to the PLC, and one-click *apply* of a fresh batch; an active run binds exactly one recipe version to one batch, giving every write a unique governance context—the window $W_r(p)$ of Eq. (1). Off-window excursions have two bounded exits: automatic rollback while the executed-rollback count is below $K=2$ (Sec. IV-B), then mandatory human escalation. Every operation lands in a central audit log with typed actor attribution (user, agent, system) and line/product/recipe dimensions; writes are also anchored in a signed journal with previous/new values, source, approval, and record identifiers—the substrate for AW-INDUSTRIALBENCH's per-write attribution metric (Sec. V) and the case study's forensic queries (Sec. VI-H).

IV. THE GOVERNED WRITE-CONTROL PIPELINE

This section formalizes the single path by which an agent actuator request becomes a plant write. A *device* is one integrated piece of plant equipment; it appears as one or more *nodes* (instrumented endpoints), and n denotes a *writable* node—one exposing a setpoint, the control image of physical quantity p . n carries a hard safety range $S(n) = [n^{\min}, n^{\max}]$ and a value transform t_n , and an active *recipe run* r prescribes a process window $W_r(p)$ for p . The accepted set is therefore the *effective constraint*

$$C(n, r) = \begin{cases} S(n) \cap W_r(p), & \text{run } r \text{ active,} \\ S(n), & \text{otherwise.} \end{cases} \quad (1)$$

Two independent mechanisms enforce containment: the interlock rejects a value inside $S(n)$ but outside $W_r(p)$, citing the window, while the driver's engineering-quantity validation rejects out-of-range values in every state. Containment thus never depends on authoring-time containment of $W_r(p) \subseteq S(n)$, and both branches of Eq. (1) are exercised dynamically in Sec. VI-C.

A. Pipeline Stages

Algorithm 1 specifies the pipeline: an *approval stage* at the tool boundary, then six ordered execution stages behind the single write entry—approval is a *decision*, execution a *validated act*. (0) *Approval gate (human in the loop)*. On a manually bound node every agent proposal is routed to a human approval request (deduplicated per node and record) and is not submitted further until a human decides; unapproved requests lapse with their record. Approval is decoupled from execution—every later stage re-runs after the decision, so a recipe change during deliberation cannot be exploited through a stale approval. The binding mode selects the posture: *manual* gates as above, *auto* skips the gate but stays subject to interlock and backstop, *unbound* has no write path at all. The gate is a property of the binding model, not a prompt. (1) *Availability gate*. A paused gateway or disabled node rejects the request: governance is unavailable exactly when actuation is unavailable. (2) *Closed-loop guardrail*. Agent writes are serialized per optimization scope—one open record per agent, records idle beyond $T_{\text{stale}} = 30$ min superseded by the next

Algorithm 1 Governed write path $\text{WRITE}(n, v, \text{src}, a)$

Require: node n , value v , source src , actor a ; active run r if any

- 1: **if** $\text{src} = \text{agent} \wedge \text{binding}(n) = \text{manual}$ **then**
- 2: route to approval; **return** pending $\triangleright$ Stage 0; lapse $\Rightarrow$ never executed
- 3: **end if**
- 4: **if** $\text{binding}(n) = \text{unbound}$ **then**
- 5: **return** reject(FORBIDDEN) $\triangleright$ tool surface never dispatches it
- 6: **end if**
- 7: **if** gateway paused $\vee \neg n.\text{enabled}$ **then**
- 8: **return** reject(CONFLICT)
- 9: **end if**
- 10: **if** $\text{src} = \text{agent}$ **then**
- 11: **if** $\exists$ open record r' of $a \wedge \text{age}(r') < T_{\text{stale}}$ **then**
- 12: **return** reject(CONFLICT)
- 13: **else if** $\exists$ open record r' **then**
- 14: $\text{SUPERSEDE}(r')$
- 15: **end if**
- 16: **if** same-direction write within T_{cd} **then**
- 17: **return** reject(CONFLICT)
- 18: **end if**
- 19: **end if**
- 20: $W \leftarrow C(n, r)$ $\triangleright$ Eq. (1)
- 21: **if** $v \notin W$ **then**
- 22: **return** reject(soft interlock)
- 23: **end if**
- 24: $(\text{raw}, \text{rb}) \leftarrow \text{driver.write}(n, t_n^{-1}(v))$
- 25: $v_{\text{rb}} \leftarrow t_n(\text{rb})$; **if** readback unavailable **then** $\text{degraded} \leftarrow \text{true}$
- 26: **if** degraded **then** $\text{ok} \leftarrow \text{false}$ **else** $\text{ok} \leftarrow |v_{\text{rb}} - v| \leq \tau(n)$ $\triangleright$ Eq. (2)
- 27: $\text{JOURNALAPPEND}(n, v^{\text{prev}}, v, \text{src}, a, r, \dots)$ $\triangleright$ executed writes only; approval and record fields elided
- 28: $\text{AUDIT}(\text{write}, n, v, a)$ $\triangleright$ 10 s dedup
- 29: **return** $(\text{ok}, v_{\text{rb}}, \text{degraded})$

writer—and within a cooldown $T_{\text{cd}} = 300$ s a same-direction rewrite is rejected while a reverse-direction correction stays allowed, so oscillatory overwrites are impossible. (3) *Soft interlock*. $v \in C(n, r)$ is checked before any driver interaction. (4) *Write with readback*. The value is encoded through t_n , sent, read back, decoded and compared under

$$\tau(n) = \frac{1}{|t_n^{\text{scale}}|} \max(0.5 \cdot 10^{-d(n)}, 0.005 (n^{\text{max}} - n^{\text{min}})), \quad (2)$$

the larger of half a least-count digit and 0.5 % of span in engineering units ($|t_n^{\text{scale}}|$: transform scale; $d(n)$: display precision). A write is ok only if the decoded readback agrees within $\tau(n)$; a driver that cannot read back yields an explicit *degraded* flag, never a silent confirmation. (5)–(6) *Journal and audit*. Every executed write appends a signed anchor ($\text{id}, \text{line}, n, \text{prev}, \text{new}, \text{src}, a, r, \text{approval}, \text{record}, t$), and the audit plane receives a typed event attributed to the actor's kind, deduplicated over 10 s.

B. Backstop and Bounded Autonomy

An optimization record freezes its pre-write baseline—the parameter value preceding the record's first governed write, with the statistics of the preceding $T_{\text{bl}} = 600$ s window—and is re-evaluated every $T_{\text{re}} = 30$ s once older than $T_{\text{win}} = 120$ s, judged against the active window $W_r(p)$ (falling back to $S(n)$, Eq. (1)) over every sample acquired since it opened. If the out-of-window count reaches $B = 3$ under an *auto* binding, the system issues a rollback verdict and executes it only while the node's executed-rollback count is below $K = 2$; otherwise the verdict is recorded and escalated to a human. The rollback target defaults to that baseline (finer granularities—stable anchor, batch start, known-good—are operator-invoked), every rollback write traverses this same pipeline, and a paused gateway defers the actuation: the fail-safe direction. Because the counter is per node and monotone, K is a node-lifetime cap.

C. Trust Boundary and Threats

The pipeline guarantees the *sanctioned* path; no framework constrains software that bypasses it. Harness engines are privileged local processes—a compromised engine could write to the protocol stacks directly, the failure mode *that* indirect prompt injection targets [35]—so the deployment model combines an availability allow-list probe (Sec. III-C), a least-privilege service account, and IEC 62443 zoning [28] to keep the governed surface the only *reachable* write path, trading zoning isolation for in-process enforceability within one zone. Inside the sanctioned path a prompt cannot *enlarge* an agent's authority; it can only abuse authority the operator already granted, which the approval gate and the chain limit bound. Governance is fail-safe under storage degradation (F6).

D. Safety Invariants

The pipeline maintains three invariants, each mapped to an implementation anchor and to the platform's scripted acceptance suite:

- **I1 (containment)**. Every executed agent write satisfies $v \in C(n, r)$ (both branches). *Anchor*: single-entry write function. *Evidence*: rejection-path acceptance tests; live-instance verification (Sec. VI-C).
- **I2 (readback closure)**. A write is reported ok only if the decoded readback lies within $\tau(n)$ or carries an explicit degraded flag. *Anchor*: Eq. (2) in the write executor. *Evidence*: deadband and calibration tests; the no-readback ablation.
- **I3 (bounded autonomy)**. Automatic rollbacks per node are at most $K = 2$ over the node's life; the third response and all manual-policy rollbacks require human confirmation, and stale records are superseded, never accumulated. *Anchor*: the per-node rollback counter guarding the backstop. *Evidence*: rollback-manager acceptance tests; the backstop drill executes one system-initiated rollback end-to-end, and the escalation-at- K branch is exercised at acceptance-test level (Sec. VI-E).

TABLE I
DATA-CREDIBILITY LAYERS; CLAIMS ARE LABELED BY THE HIGHEST LAYER MEASURED.

Layer	Description	Status
L1	Seeded simulation: two independently defined plant scenarios (strategy-signal film line; ODE cast-film twin), seeded noise, offline optimum W^*	measured
L2	TEP [21] / SWaT [22] replay through real drivers	designed
L3	One physical OPC UA/Modbus device on the real stack; write-readback latency	designed
L4	Digital-twin case study: seeded closed-loop campaign on the cast-film line—deterministic controller over three seeds and an agent team over four runs (Secs. VI-F, VI-H)	measured

These are stated over the pipeline, not over the LLM: whatever the agent proposes, the accepted-action language is constrained. This is the precise sense in which the system’s autonomy is *governed*.

V. AW-INDUSTRIALBENCH: BENCHMARK AND EXPERIMENTAL PROTOCOL

A. Design Principles and Data Layers

AW-INDUSTRIALBENCH is built on three principles. *Determinism*: a seeded PRNG (mulberry32) and a scenario-snapshot API make structural metrics (interception verdicts, judge-class checks) point-wise reproducible; objective values over *sampled* process variables remain timing-sensitive (Sec. VI-I); the offline grid-search optimum W^* is the reference bound. *Layered realism*: results are reported per data layer (Table I), never pooled, so claims carry their realism level. *Executable protocol*: the released audit harness (bench/run.mjs; four seeded tiers) executes the artifact and invariant checks of Secs. VI-C–VI-C; the E1–E9 campaign configurations are declarative documents (E-numbers instantiate the B groups; E8 = B4).

B. Tasks, Faults, Metrics, and Baselines

Five task families cover the supervisory workload, each judged by a machine-readable oracle over the database, read-back values, and event streams, never log strings: *T1* closed-loop setpoint correction in-window; *T2* out-of-window excursion; *T3* multi-node joint adjustment under the step rule; *T4* batch parameter recovery; *T5* mid-run objective change and re-planning. Six fault families inject at the simulator control API: *F1* step drift, *F2* sensor freeze, *F3* communication interruption (30 s), *F4* out-of-order injection, *F5* out-of-window writes, *F6* storage unavailability—*F5* stresses governance (I1), *F6* degraded-mode integrity.

Metrics form three families. *Safety* metrics require the governed pipeline: out-of-window write interception rate (denominator = seeded attacks), readback-failure capture rate, false-block rate (denominator = legal in-window writes attempted), spurious-rollback rate, and per-write attribution rate (target 100%). *Performance* metrics cover the *automatic segment* (proposal acceptance to verified write), kept separate from human-approval waiting time so human response is never

conflated with system latency. *Cost/quality* metrics cover task goal rate, tokens and LLM calls per task, approval wait time, and human interventions per task.

Eight baseline groups are defined. B1 ablates the pipeline’s own mechanisms (bench-mode-only bypass switches). B2–B3 isolate the platform from its intelligence: they run general frameworks (AutoGen [23], LangGraph [24], CrewAI [25]) and ReAct-style prompting [26] on the same MCP tool surface [27]. B4 is a human-operator reference under the τ -bench paradigm [13]; B5 anchors against classical control with W^* as reference; B6–B7 are the scheduling and memory arms of Sec. III-C; B8 replicates tasks across real engines.

Reproducibility protocol. Released today: the harness executes every check of Secs. VI-C–VI-C with fixed seeds; every run records its seed, harness source hash, and commit, appends to never-overwritten timestamped directories, and yields judge-class metrics identical across executions (latency is environment-dependent and reported as ranges). *Specified, in development*: the campaign runner, the statistics module (Mann–Whitney U with Holm correction, bootstrap 95 % CIs; unit of analysis = run), the fault injector, and the decision-log instrumentation persisting each LLM decision’s prompt hash, model version, and token usage. Released LLM arms pin model versions; temperature-0 applies to the campaign configurations. PolyForm Noncommercial permits academic reproduction; commercial reuse is a licensing boundary, not a reproducibility one.

VI. EXPERIMENTAL EVALUATION

A. Setup

All results were produced by one executable pipeline (bench/pipeline.mjs) that boots the plant simulator and the platform, provisions the plant over real protocol transports, drives the governance surface, runs the closed-loop campaign, and writes a machine-readable archive; one command reproduces the section:

```
node bench/pipeline.mjs --profile integrated --seed
42 --cl-seeds 3
```

Every reported number is read from a never-overwritten run directory recording the seed, the harness source hash (SHA-256 over the eight checker sources), the commit, the platform, and the evidence chain. The integrated runs quoted here are the 2026-09-14 pair (20260914152838-fwg, hash 204fc9ee) and its reproduction twin 20260914153400-ce0, unless a range over all released runs is stated. Host environment: an AMD Ryzen 7 9700X (8 cores) host with 31 GB RAM running Windows. Environment-class quantities (latency, sampling counts, wall time) vary across releases and are reported as archived ranges; judge-class quantities (interception, false blocks, boundaries, attribution, structural verdicts) were bit-identical in every same-seed re-run.

The fourteen stages (P0–P9 with P4b–P4e and P8b; no P5) run preflight and bootstrap, plant instantiation with the offline optimum, multi-line provisioning over five drivers, acquisition and interlock checks, the deterministic tool-level control loop over three observe–actuate–judge cycles per line, the

approval gate, audit reads, the recipe lifecycle, the closed-loop benchmark, multi-modal acquisition, cross-scenario portability (Sec. VI-B), the backstop drill and the subsystem benchmarks (Sec. VI-E). A full run over five devices takes about 5.2 min—the closed-loop benchmark about 36 s per plant seed, the backstop drill a cadence-bound 120–130 s window.

Headline results. Across both scenarios the pipeline intercepted 33/33 seeded out-of-window/range writes (Wilson 95 % [89.6 %, 100 %], zero false blocks)—24 real-protocol attacks (Sec. VI-D) plus 9 portability attacks (Sec. VI-B); the six API-tier probes of Sec. VI-C re-exercise the same attacks and are not double-counted—recommissioned the deployment onto the second scenario with zero platform code changes, recovered 96.5–97.3 % of the offline grid optimum in the governed closed loop under a fixed controller (quoted run: 97.0–97.2 %), and exercised the backstop end-to-end (≈ 120 –130 s). An LLM agent team closed the same loop over real transports through the same path, but reached only a median 0.76 of that optimum (Sec. VI-H).

Scope. The reported run exercises T1–T2 and F2/F3/F5 against baseline B1; T3–T5, F1/F4/F6, hardware-in-the-loop, public-dataset replay, cross-engine replication, and B2–B8 are covered by the released campaign designs and are *not* reported here. All measured layers are simulation-tier (Table I).

B. Cross-Scenario Portability (Framework Claim)

The central framework claim—adapting the deployment to a new line is a configuration task, not an integration project—is measured directly by the pipeline’s final stage. Everything is first provisioned on the *cast-film physics* scenario (ODE plant model, six actuators, seven sensors, five protocol drivers, vector and image frames); the plant then switches to a second, independently defined scenario, *film-line*: a coating-oven line with a disjoint signal domain and per-signal strategies rather than a physics engine. No scenario-specific code exists in the commissioning path—the identical five-step flow (export, connection test, node creation, line/product/recipe provisioning, agent binding) re-instantiates the deployment on both, because driver configurations derive from each device’s export (Sec. III-A, P3).

Measured on the second scenario: five devices across all five protocol drivers are commissioned in one pass (the commissioning artifact is five derived driver configurations plus one scenario identifier); three writable-setpoint devices become governed lines (Modbus TCP, OPC UA, MQTT) and the two acquisition-only devices attach as satellites; all five nodes stream live data (8–12 one-second buckets in the quoted pair); governed writes are accepted with readback deviation $\leq 0.04^\circ\text{C}$ on Modbus TCP and OPC UA (exactly zero on OPC UA in the quoted run); and all nine seeded out-of-window attacks are rejected with zero false blocks (none of the 3 legal in-window writes blocked). The stage completes in about 16 s in the quoted run (15.7–23.0 s across releases) and the rig is restored. The structural verdicts—100 % interception, 0 % false blocks—are identical across scenarios: per-line engineering moved into authoring the scenario definition, and the platform absorbed the difference unchanged.

Fig. 2 shows that path as an operator drives it on the live instance: provisioning a device as a *node* whose driver, cadence and line derive from its export (1); binding it to a device twin and a recipe window, so the process window travels with the deployment (2); deploying one lead plus harness-backed workers into a channel, where one team definition runs on an LLM engine or on the deterministic replay engine (3); and each write then opening an optimization record that *dcw_judge* resolves to *keep* or *rollback* beside the setpoint ledger (4). No plant-specific code appears in any of the four surfaces: this is the integration claim of Sec. III made operational.

C. Static Audit and Dynamic Invariant Verification on a Live Instance (L1 Tier)

Before any dynamic campaign, the released audit harness (*bench/run.mjs*) verifies the artifact at source level. Three checks run offline: a *capability inventory* against the claims of Sec. III; a *paper-code consistency* check that locates 20 representative quantitative constants cited in this paper (sweep quota 64, breach threshold 3, rollback-chain limit 2, RRF $K=60$, MMR $\lambda=0.7$, compaction 0.7, watchdog 300 s, hysteresis 2 %, among others) at their implementation anchors; and a *pipeline stage-ordering* check asserting that the governed-write stages of Algorithm 1 appear in the declared order at their source anchors. All pass: 20/20 anchors and 2/2 ordering chains—engineering discipline rather than validation evidence, since the harness audits the artifact of which it is part.

The static audit verifies structure; the rest of this subsection verifies behavior: L1-tier evidence (Table I), driving a pinned platform instance (v0.7.39, mock drivers) exclusively through its REST API with seeded randomness (*mulberry32*, seed 42). A benchmark line is created with global safety range $[120, 260]^\circ\text{C}$ and a batch started with active recipe window $[175, 205]^\circ\text{C}$; the safety invariants of Sec. IV-D are then attacked and probed, including the decision boundaries and both branches of Eq. (1).

I1 (containment). Six seeded gross out-of-constraint writes—global-range violations (107, 269), window violations while the batch is active (219, 163), and extreme/negative values (10^6 , -25)—are all rejected with typed validation errors, while all three in-window control writes are accepted: 100 % interception at 0 % false blocks. Boundary probes at $\pm$ one least-count behave as specified (205.0 accepted; 205.1, 174.9 rejected), and the rejection message cites the window as the binding constraint. With the batch stopped the window branch deactivates and the hard range alone governs: an out-of-range write (300) remains rejected while a within-range, out-of-window write (130) is accepted—the soft interlock is a batch-scoped constraint, not a constant.

I2 (readback closure) and attribution. All accepted writes return a driver readback within the tolerance of Eq. (2) ($\tau = 0.7^\circ\text{C}$ at one decimal digit and a 140°C span) and produce journal anchors with source, actor, timestamp, and previous/new values—a 100 % attribution rate. Acquisition held the 1 s cadence, and the samples API answered 20 probes at p50 6–22 ms / p95 8–63 ms.

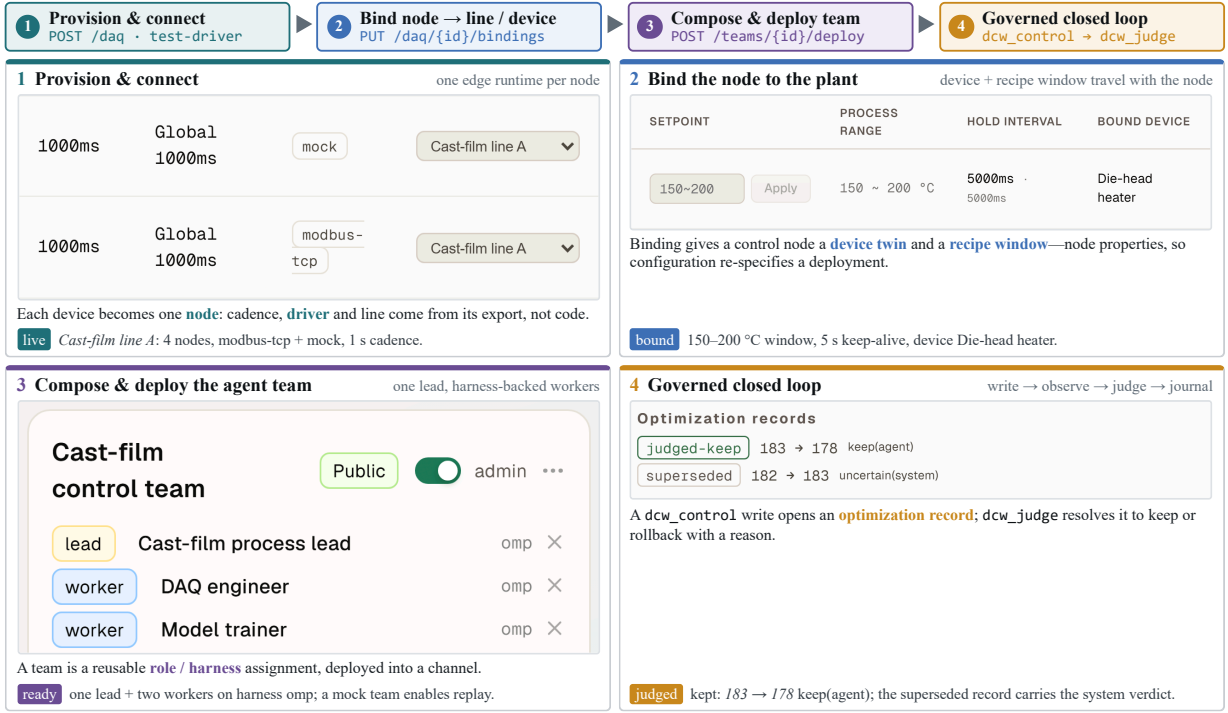


Fig. 2. Operational walkthrough captured on the live instance (seeded plants). A device becomes a *node* whose driver and cadence are configuration (1); the node is *bound* to a device twin and a recipe window (2); one lead plus harness-backed workers deploy into a channel (3); and every subsequent agent write opens an optimization record that `dcw_judge` resolves to *keep* or *rollback*, shown against the setpoint ledger (4). No surface contains scenario-specific code.

Reproducibility. Every execution on a pinned harness—the two identical-harness pipeline runs of Sec. VI-I and the released ablation runs (three aggregated in Sec. VI-G)—produced identical judge-class metrics: the same seeded attacks, rejections, boundary verdicts and ledger outcomes; only fixture identifiers and latency varied. `bench/compare.mjs` asserts judge-class equality between a fresh run’s record and the archived baseline (REPRODUCIBLE; the gate rejects a tampered metric, validated by a self-test). The Wilson 95 % interval for 6/6 trials ([61 %, 100 %]) is the appropriate reading of any small-*n* rate.

Scope. These checks exercise the accepted-action language at the API boundary with the LLM out of the loop—deliberately so, since the invariants are defined over the pipeline, not over any engine (Sec. III-A, P4).

D. Real-Protocol Plant Scenarios (Five-Protocol PLC Node Simulator)

The API-tier checks of Secs. VI-C and VI-G use in-process mock drivers; this subsection re-runs the working-condition scenarios over *real protocol stacks* against an independent five-protocol PLC node simulator (Modbus TCP/RTU slaves, OPC UA server, MQTT broker/publisher, HTTP endpoints, ODE plant model). Driver configurations derive at runtime from the simulator’s integration-export API—no hand-written connection settings—and every candidate driver passes the connection test before a node is created.

The pipeline provisions one production line per device that exposes a writable setpoint (4 lines: Modbus TCP, OPC

UA, MQTT, HTTP) and attaches the acquisition-only device (a Modbus RTU gel counter) as a *satellite* node on the first line, because acquisition samples only nodes whose line has an active batch. All five nodes stream live data through their real transports (2–14 one-second buckets in the settle window, Table II). A governed setpoint write costs 19–112 ms (p50) by protocol, the Modbus TCP transaction dominating (its p95 reflects the first write’s connection setup); readback closes within 0.04 °C on Modbus TCP and exactly on OPC UA, while the MQTT and HTTP control channels report no readback by design (publish-only / echo-only actuators)—a driver property the write path records rather than conceals. Six seeded attacks are rejected on *every* writable protocol (6/6 each; 24/24 pooled, Wilson 95% interval [86 %, 100 %]) with zero false blocks on the four legal in-window writes, and the deterministic tool-level control loop (`daq_query`→`dcw_control`→`dcw_judge`) converges on all four lines. A link-interruption drill (simulator service stopped and restarted) is detected and recovered, and a process-freeze injection shows defense in depth: the governed write still passes (readback verifies the register, not the process) while the independent recipe-window monitor catches the failure within 2 s—the layer separation of Secs. VI-C and IV-D.

E. Governance Surface: Recipe Lifecycle, Rollback, and Approval

With the LLM out of the loop, every probe below ran end-to-end in one pipeline run and exercised the *lifecycle objects* the governance stack manages. *Non-destructive history*:

TABLE II

REAL-PROTOCOL SCENARIOS OVER THE FIVE-PROTOCOL PLC NODE SIMULATOR (L1, SEED 42). FOUR LINES PLUS ONE ACQUISITION-ONLY SATELLITE. ACQ. PTS: SAMPLES IN THE SETTLE WINDOW. WRITE LATENCY: P50/P95 OVER SIX IN-WINDOW WRITES PER LINE. “N/A”: PUBLISH-ONLY (MQTT) / ECHO-ONLY (HTTP) ACTUATORS REPORT NO READBACK. F5: SEEDED OUT-OF-WINDOW WRITES REJECTED, OF 6 PER PROTOCOL.

Protocol	Acq. pts	Write p50/p95 (ms)	Readback	F5
Modbus TCP	2	111.5 / 141.6	0.04 °C	6/6
OPC UA	11	19.0 / 22.1	exact	6/6
MQTT	12	19.4 / 19.5	n/a	6/6
HTTP	12	32.8 / 33.2	n/a	6/6
Modbus RTU [†]	14	—	—	—

[†]acquisition-only satellite node (gel counter).

a revert *creates* version 3 rather than overwriting. *Judge-execute separation*: judging `rollback` changed no PLC value (196.60 °C throughout)—only the separately invoked `rollback` restored the origin value (203.40 °C); an agent’s verdict is an intent record, not an actuation. *Bounded autonomy in action (backstop drill)*: freezing the process variable out of window under an open auto-binding record triggers the backstop—the record is judged `rollback by the system` and the PLC restored to the baseline (186.2 ± 0.75 °C), cadence-bound at 120–130 s across releases; chain limit $K=2$ remains specified (Sec. VII-A). *Default non-execution*: in manual mode an agent write blocks until the approval releases it (148.2 rpm on OPC UA in the quoted run; poll-inclusive decision latency ranged 6–468 ms across released runs); the journal holds ten anchors over four sources and both log surfaces return full 200-entry windows.

Platform subsystems (P7/P9): vector and image frames from the profile and CCD channels are stored as first-class acquisition objects (P7); a mock lead dispatches a child task that a worker completes (parent COMPLETED); duplicate team-memory writes under one deduplication key yield one entry; the registry enumerates 14 engines, with environment-available counts of 12–14 across releases (one run’s probe glitched to zero and is excluded).

F. Closed-Loop Optimization Benchmark

The closed-loop benchmark quantifies how much of the offline optimum a *governed* loop recovers on the cast-film twin. The pipeline instantiates all six actuators (zones on Modbus TCP, screw on OPC UA, line speed on MQTT, die gap on HTTP), starts a batch off-optimum ($N=150$ rpm, $v=95$ m/min, zone 200 °C), and runs a controller that reads through the acquisition path and writes through the governed path. The objective

$$J = 55 J_{\text{thick}} + 25 J_{\text{quality}} + 8 J_{\text{energy}} + 7 J_{\text{throughput}} \quad (3)$$

constrains melt temperature to [195, 225] °C and pressure to ≤ 22 MPa.

The controller is model-light and fixed in advance: it inverts $h \propto N/v$ at constant line speed ($N \leftarrow N \cdot (50/h)$, ± 22 rpm per step) and corrects the heater setpoint only when melt temperature leaves [196, 224] °C. It contains no LLM,

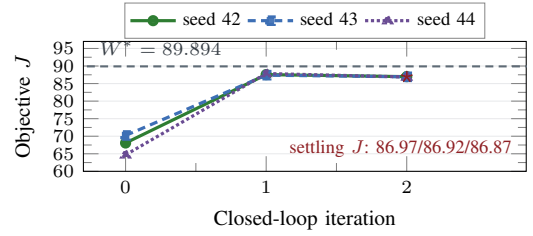


Fig. 3. Governed closed-loop optimization on the cast-film twin (three seeds, real transports, deterministic controller). J is the objective of Eq. 3; all seeds converge in two iterations to 97.0–97.2% of the offline grid optimum W^* (dashed). Red stars mark the iteration-2 settling measurement; the quoted ratios are the J_{end} means of both evaluations over W^* .

so the benchmark isolates the *governed loop* rather than a model. Convergence requires two consecutive evaluations with $h \in [48, 52] \mu\text{m}$, defect $< 2\%$, pressure ≤ 22 MPa.

Fig. 3 reports the trajectories; J_{end} is the mean of the two consecutive in-band evaluations. From $J_0 = 64.7\text{--}70.2$ the loop reaches $J_{\text{end}} = 87.2\text{--}87.4$ against the offline grid optimum $W^* = 89.894$ (weights unnormalized, 0–95 scale; real-time physics), i.e. 97.0–97.2% of the optimum in the quoted run, converging in two iterations with at most two governed writes per seed, no rejected write and no constraint violation; mean wall time is 36 s per seed. Thickness moves from 55.2–56.1 μm (target $50 \pm 2 \mu\text{m}$), defect settles at or below 0.64%, pressure at 16.6–16.9 MPa, and melt temperature stays inside [195, 225] °C. Per-seed values in the quoted run ($J_0 \rightarrow J_{\text{end}}$, ratio): seed 42: 68.07→87.26 (0.971); seed 43: 70.21→87.16 (0.970); seed 44: 64.69→87.37 (0.972). Across the 25 completed releases of the same command and seed (three early-development and two interrupted runs excluded), per-seed J/J^* spans 0.965–0.973 and run-level means 0.968–0.971; the residual spread reflects the sampled process variables from which J is computed. The 3% shortfall is a throughput/energy trade, not a control failure: the grid optimum sits at a low-throughput corner where the objective’s energy weight slightly exceeds its throughput weight, while the loop holds the higher line speed it started from.

G. Ela: Pipeline-Tier Governance Ablation (Completed)

Using the env-gated bench-mode bypass (production builds refuse the flag), we compare the full pipeline against the three ablation arms of B1—soft interlock bypassed (no-interlock), readback verification bypassed (no-readback), and fully ungated—on the live instance (mock drivers, seed 42, three repetitions per arm per released run, fresh fixture per repetition). Because the hard range is enforced structurally at the driver layer and cannot be bypassed, the ablation isolates exactly the two bypassable mechanisms. Fig. 4 reports the result; values were identical across repetitions. Three findings: (i) bypassing the interlock collapses interception from 6/6 to 4/6 (identically in the no-interlock and ungated arms)—the window-class attacks execute while global-range attacks remain structurally rejected; (ii) the executed breaches are fully journaled (twelve out-of-window anchors per bypass arm per run: six breach writes plus

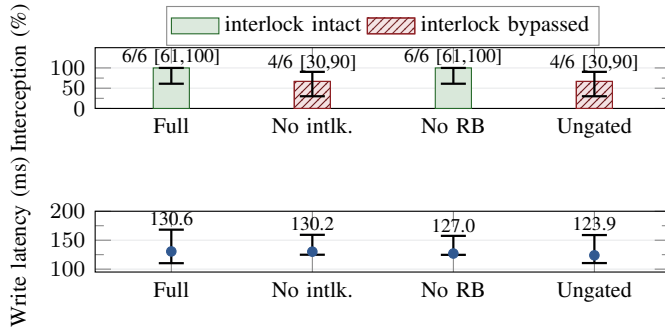


Fig. 4. Elia four-arm governance ablation (mock drivers, seed 42; nine repetitions per arm across three released runs, judge-class values identical). Top: interception per arm; error bars are Wilson 95% intervals over the six seeded attacks, with endpoints printed—structural rather than statistical at $n=6$. Bottom: mean p50 write latency per arm as point ranges, not bars, so a truncated axis cannot exaggerate differences; whiskers span the minimum observed p50 to the maximum observed p95, and arm means are printed. Aggregates recomputed from the released per-repetition CSVs by `bench/tools/fig4_analyze.py`.

six accepted edge probes): *attribution is not prevention*; (iii) no arm adds latency resolvable at this tier (p50 110–141 ms, p95 150–169 ms across all arms). Boundary behavior follows each arm’s specification (9/9 probes; Wilson 95% [70, 100]%), false blocks are 0/3 in every arm, and the no-readback arm matches Full on in-memory drivers (the mock readback echoes the command); its verification value is exercised by the frozen-process injection above.

H. Case Study: LLM-Agent Closed-Loop Optimization (L4)

The benchmark of Sec. VI-F isolates the governed loop under a fixed controller; this subsection reports the complementary result—the same plant and objective closed by an LLM agent team. Four campaigns are released (first-party omp engine driving glm-5.3-flash in a lead-supervised team; manual-mode bindings; physics accelerated 6 $\times$; seed 42; plant and objective as in Sec. VI-F). Their sampling settings predate the temperature-0/model-pinning discipline of Sec. V (harness defaults), one reason we report them as an existence proof rather than a controlled measurement. In each campaign the team opened an optimization record, probed the plant with `daq_query`, and actuated through `dcw_control` over real protocol transports (Modbus TCP, OPC UA, MQTT, HTTP); the platform’s write ledger records 7, 7, 15 and 10 governed writes respectively with per-write journal anchors held in the platform archive, and approval decisions recorded by the campaign’s scripted operator emulation under the campaign operator identity—an autonomous-campaign configuration, not a production posture.

The outcome is unfavorable to the agent. The agent objective reached $J_{\text{agent}} = 67.7, 31.2, 68.3, 68.5$ against $W^* = 89.894$ —0.75, 0.35, 0.76, 0.76 of the offline optimum, median 0.76—and the thickness window $50 \pm 2 \mu\text{m}$ was *not* reached: three campaigns settled at a $55.33 \mu\text{m}$ window mean and the fourth produced no in-window samples (its MQTT thickness channel also refused connection), while the melt-pressure node

failed its connection test on OPC UA in all four. Of 39 acceptance checks per campaign, 33–35 passed.

We report this as a negative result with a design implication. It supplies the existence proof the framework needs—an LLM team drove real protocol transports through the governed path with every write reconstructable—while showing that on this objective the team does not match a two-step fixed controller (0.76 vs 0.97 of W^*). That is the propose/dispose division of roles working as intended: the agent explores and proposes, and the deterministic path converts proposals into converged actuation.

I. Reproducibility and Threats to Validity

Re-running the whole pipeline twice with an identical harness (source hash 204fc9ee; runs 20260914152838-fwg and 20260914153400-ce0), seed, and commit yields *bit-identical* structural results: 55/55 pipeline checks pass in both runs; every seeded attack is rejected with zero false blocks on *both* plant scenarios (9/9 on the second); all four per-line loops converge; the portability and backstop-drill outcomes are identical; and $W^* = 89.894$. The one structural difference is benign: governed-write totals are 6 vs. 5 because seed 44 converged on its first write—“at most two governed writes per seed” still holds. What varies is what the abstraction predicts: governed-write latency is environment-dependent, and the closed-loop objective—computed from *sampled* process variables—lands in $J/J^* \in [0.965, 0.973]$ across campaign repetitions. Every run appends to a never-overwritten timestamped directory; the archive holds more than 90 run records across tiers, each with its seed, harness hash, and commit.

Threats to validity. Construct: success oracles are database- and readback-defined, mitigating self-reported success bias. *Internal:* temperature-0, pinned-version LLM arms reduce but do not eliminate variance; hence mock-arm matrices with LLM spot-checks and nonparametric tests. *External:* all measured layers are simulation-tier (two seeded plant scenarios, bounded by the ODE twin’s fidelity); public-dataset replay (TEP, SWaT) is designed but unmeasured; the four LLM campaigns share one seed and carry no cross-seed variance estimate. Results are layer-labeled (Table I) and we claim no transfer beyond L1–L3–L4 without per-layer evidence. *Conclusion:* the closed-loop result is reported over a fixed, pre-declared controller and three plant seeds, and the LLM campaign is reported as a measured negative rather than a headline.

VII. DISCUSSION AND LIMITATIONS

A. Scope and Limits

The pipeline constrains the *action language* of agents, not the quality of their choices: proposals still derive from LLM competence, context quality, and tool ergonomics. Bounded mechanisms (interlock, readback, chain limit) absorb bad proposals; the unbounded statistical components (LLM decisions, with deterministic rule fallbacks) bottom out at the rules’ behavior. The backstop’s $K=2$ caps a node’s lifetime automatic rollbacks—an engineering limit exercised only up to one in the drill. The invariants borrow IEC 61511’s separation of basic

control from safety functions [29] but remain engineering properties, not certified safety functions: the by-construction guarantee holds within the sanctioned path of Sec. IV-D; AGENTWORKSHOP does not replace certified instrumentation. The completed evidence covers the artifact audit, live-instance invariant verification, a dynamically exercised rollback, real-protocol scenarios on both plant scenarios, the pipeline-tier ablation, and one plant domain at L1/L4; the remaining agent-in-the-loop arms (B2–B8) are released designs rather than reported measurements—AW-INDUSTRIALBENCH measures *governed supervisory control*, not open-ended plant engineering. Three lessons may transfer: single-entry writes are the keystone; judgment/execution separation pays twice; deterministic rehearsal against an offline-optimum oracle surfaces defects before hardware.

VIII. CONCLUSION

We presented AGENTWORKSHOP, an integration framework embedding LLM multi-agent teams natively into the supervisory fabric of protocol-heterogeneous production lines—one process, one data domain, physical semantics instead of register addresses, team-level node binding, configuration-only commissioning—with AW-INDUSTRIALBENCH, a reproducible evaluation protocol where every agent-issued write passes through a single governed pipeline defining three safety invariants over the accepted-action language, independent of engine. A governed closed-loop benchmark on a seeded cast-film line recovers 97.0–97.2 % (quoted run) of the offline grid optimum in two iterations and at most two governed writes per seed (per-seed $J/J^* \in [0.965, 0.973]$ across released runs); the backstop’s rollback is exercised end-to-end on a live breach; and four LLM-agent campaigns actuate the same loop over real transports with every write attributed, reaching a median 0.76 of the optimum—a shortfall that motivates the propose/dispose split. The remaining campaigns of AW-INDUSTRIALBENCH are released as executable protocol designs; the platform source (PolyForm Noncommercial), the benchmark harness with frozen baselines and run archives, the seeded simulator, and the PLC node simulator (git submodule) are versioned in the project repository.

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